

Validation Roadmap for the Prior-Informed Bootstrap Framework

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Companion document to the Framework Specification

For developer and collaborator review

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Abstract

This document specifies the validation strategy for the prior-informed bootstrap framework introduced in the companion specification. It defines the network corpus on which the framework will be tested, the missingness mechanisms used to degrade fully observed networks into incomplete observations, the comparison baselines, the evaluation criteria, and the success conditions that determine whether the framework is ready for publication and broader use. The validation design draws methodologically from [Smith et al. \(2022\)](#), adapting their Monte Carlo protocol for the framework’s distinctive credible-interval output. The document is intended as a commitment device: the conditions and criteria specified here are the ones the framework will be evaluated against, settled before implementation begins.

1 Introduction and Scope

The framework specified in the companion document introduces several methodological innovations relative to prior bootstrap approaches for network credible intervals — explicit node-level missingness, a user-supplied prior on centrality-missingness correlation, structural-role binning via the E/I index, and rank-conditional reciprocity imputation for nominated non-respondents. Each of these innovations expands the framework’s applicability beyond what existing methods can handle, but also introduces additional design choices and assumptions that must be validated empirically before the framework can be recommended for general use.

This document specifies how that validation will be conducted. It defines:

- A network corpus combining empirical and synthetic networks spanning size, directedness, and centralization;
- Procedures for degrading fully observed networks into incomplete observations under controlled missingness mechanisms;
- Comparison baselines selected to represent the realistic alternatives a practitioner would consider;
- Evaluation criteria that capture both the framework’s Bayesian credible-interval output and its compatibility with point-estimate comparisons established in prior work;
- Concrete success conditions against which the framework’s performance will be judged.

The validation explicitly does not aim to dominate every existing imputation method in every setting. Prior work ([Smith et al., 2022](#)) has established that no single imputation approach is uniformly best across networks, measures, and missingness mechanisms. The framework’s value proposition is providing well-calibrated

credible intervals under non-random missingness, a capability that prior simple-imputation methods do not offer at all. The validation is therefore structured to demonstrate three claims: (1) the framework provides credible intervals with reasonable coverage and useful width across a range of realistic conditions; (2) the framework’s point estimates are competitive with simple imputation methods on the metrics those methods were designed to optimize; and (3) the framework’s calibration degrades gracefully when the user’s supplied prior departs from the true missingness mechanism.

2 Validation Philosophy

The validation rests on four principles.

Treat empirical networks as ground truth. For each empirical network in the corpus, the network as collected is treated as the “true” network for evaluation purposes. We do not attempt to correct for any measurement error or missingness present in the original data collection; rather, we induce additional, controlled missingness on top of the original data and measure how well the framework recovers the original. This is the standard Monte Carlo paradigm in the network sampling literature (see [Smith et al., 2022](#)) and produces results that are interpretable as “how well does the framework recover the network the analyst would have if they had complete data?”

Use synthetic networks for controlled conditions. The empirical networks span specific structural cells but cannot be modified to test how performance varies along a single structural axis. Synthetic networks let us hold most properties constant and vary one at a time — size, centralization, density, community structure. The synthetic networks are not meant to replace the empirical ones; they complement them by providing controlled stress tests.

Match the methodological precedent where possible. Where [Smith et al. \(2022\)](#) established procedures for inducing missingness and evaluating bias, this validation follows those procedures. This produces results directly comparable to that prior work and avoids re-litigating methodological choices that have already been validated in the literature. Where the framework’s distinctive features require new evaluation procedures (e.g., for credible-interval coverage), this document specifies those new procedures explicitly.

Validate the reconstruction directly at the network level. Beyond the measure-level coverage and bias tracks, the reconstruction step is tested in its own right: for each network we ask whether it makes the observed network look more like the truth, comparing how closely the reconstructed networks resemble the true network T against how closely the raw degraded network G_{obs} does. The degraded network is the reference — reconstruction earns its keep only if it moves G_{obs} closer to T than leaving it untouched would — and a single network-similarity score, applied identically to both sides, isolates the gain. This check is run at the MCAR condition ($\rho = 0$), where no achievable- ρ ceiling constrains the design, so every network and missingness rate is in scope and the comparison covers the full corpus.

3 Network Corpus

The corpus comprises sixteen networks spanning the four cells of the weighted $\times$ directed design: five empirical, four synthetic at base specification, two thresholded unweighted variants of empirical networks, and four unweighted synthetic networks. The empirical networks cover real-world structural complexity at sizes ranging from 70 to 6,486 nodes. The synthetic networks fill structural cells the empirical corpus

does not cover and provide controlled-condition baselines. The unweighted variants ensure the framework’s distinctive ability to handle unweighted networks — absent in [Bellutta \(2026\)](#)’s Chapter 4 framework — is exercised across both directed and undirected settings. The Marvel co-appearance network ($N = 6,486$) was added as a large undirected representative, extending the corpus’s size range by nearly a factor of five beyond the next-largest network and providing a large-and-clustered undirected counterpart to the large-and-sparse directed Balikpapan case.

3.1 Empirical Networks

Moreno High School (directed, $N = 70$). Friendship nominations among boys in an Illinois high school, aggregated across fall 1957 and spring 1958 surveys. Edge weights range 1–2 indicating whether each boy was nominated in one or both surveys. Density 7.6%, mean degree 10.5, reciprocity 0.50, clustering coefficient 0.40. The network is small, dense, and highly reciprocated, with moderate clustering. It represents the canonical case where simple probabilistic imputation (using reciprocity as the imputation signal) is expected to work well, and serves as a natural test case for the framework’s rank-conditional reciprocity matrix R .

Toledo Crime Networks (undirected, $N = 77$). Multiplex criminal cooperation network covering seven crime types: drug trafficking, theft, weapon carrying, homicide, kidnapping, arms trafficking, and ideological falsehood ([Toledo et al., 2023](#)). For this validation, the seven layers are collapsed to a single weighted summary network via sum-across-layers, following the procedure used in [Bellutta \(2026\)](#) for evaluating single-layer methods on multiplex data. The collapsed network has 223 weighted edges and density approximately 7–8%. Provides a small undirected empirical case with multiplex structural origin.

Scotland Interlock (undirected, $N = 108$). Bipartite director-company affiliation network for Scottish joint stock companies in 1904–1905, derived from [Scott and Hughes \(1980\)](#). For this validation, the bipartite network is projected to a one-mode company network: two companies are connected if they share one or more directors, weighted by the number of shared directors. The projection produces 108 nodes with company-level attributes (industry category, total capital). Provides an empirical case with known community structure (industry categories) that can serve as ground truth for Louvain community detection validation.

Balikatan 2022 (directed, $N = 1,347$). Twitter communication network from the Balikpapan 2022 US-Philippines military exercise. Edges aggregate mentions, retweets, replies, and quotes between accounts; edge weights are integer interaction counts. Density approximately 0.17% (sparse), transitivity 0.23, modularity (Leiden) 0.56. Provides the corpus’s only large, sparse empirical case with strong community structure. Most relevant for testing the framework’s scalability and the value of E/I binning in a setting where community structure is strong but the network is too large for model-based imputation.

Marvel Universe (undirected, $N = 6,486$). Character collaboration network derived from [Alberich et al. \(2002\)](#)’s compilation of Marvel comic book data. The source is a bipartite hero–comic graph (6,486 heroes, 12,942 comics, $\sim 96,663$ hero–comic incidences), projected to a one-mode hero–hero network via BB^T where two heroes are connected with weight equal to the number of comics in which they jointly appeared. Pairs sharing only one comic are dropped via a threshold of co-appearance count ≥ 2 , retaining 77,227 collaboration edges; this threshold removes the noisy long tail of incidental single-comic co-appearances while preserving the substantive collaboration structure. The resulting weighted undirected network has mean degree 11.9, transitivity 0.79, and a heavy-tailed weight distribution (weights range 2–744, Gini 0.54). The largest connected component spans 70% of nodes. Marvel is the corpus’s large-undirected representative: it

stress-tests the framework’s scalability on a network nearly five times the size of Balikpapan, in the undirected-and-clustered regime that complements Balikpapan’s directed-and-sparse one. Its heavy-tailed degree and weight distributions also make it the most demanding case for the weighted path-based measures and the layered triad census. Its centrality distribution, with a small number of high-degree hubs and a long thin tail of heroes at degree 1 or 2, produces asymmetric achievable- ρ ceilings between positive and negative directions in the degeneration grid (Section 4).

3.2 Synthetic Networks

The synthetic networks are designed in pairs, with each pair sharing an underlying structure but generated in both directed and undirected variants. This produces a clean experimental contrast between directionality holding other properties constant. Synthetic networks are generated with controlled parameters but realistic structural properties; specific parameter values may be refined after preliminary analysis of the empirical corpus (see Section 3.5).

Synthetic 1 (low centralization, $N = 600$). Stochastic block model with 4–6 communities of approximately equal size and moderate inter-community connectivity. Target density 5–7%, target clustering coefficient 0.30–0.40, target in-degree centralization 0.05–0.10. Edge weights drawn from Poisson($\lambda = 2$). Generated as a directed network; the undirected variant is produced by symmetrizing (union of forward and reverse edges, with weights summed). Provides a controlled large network where community structure is known by construction (allowing direct validation of Louvain recovery) and centralization is low (testing the framework where bin-exchangeability should hold most cleanly).

Synthetic 2 (high centralization, $N = 250$). Directed preferential attachment with attachment proportional to in-degree only, producing extreme in-degree concentration with uniform out-degree. Target density 2–3%, target in-degree centralization 0.25–0.30, target Bonacich centralization 0.35–0.45, target reciprocity 0.10–0.20 (low). Edge weights drawn from Poisson($\lambda = 2$). Undirected variant produced by symmetrization. Provides the corpus’s controlled high-centralization case — the structural opposite of Moreno — and a stress test for bin-exchangeability when degree distribution is highly skewed.

3.3 Unweighted Variants

The framework’s ability to handle unweighted networks is one of its distinctive capabilities relative to [Bellutta \(2026\)](#)’s Chapter 4 framework. To validate this capability across the four cells of the weighted $\times$ directed design, unweighted variants of four networks are added to the corpus, providing at least three networks in each cell.

Moreno (unweighted directed). Produced by thresholding the weighted Moreno network: any directed pair with weight ≥ 1 becomes an unweighted edge. The resulting network has the same edge presence pattern as the weighted version but loses the distinction between single and double nominations.

Toledo (unweighted undirected). Produced by thresholding the collapsed weighted summary: any unordered pair with at least one observed cooperation across the seven crime layers becomes an unweighted edge. The unweighted variant arguably aligns more closely with the underlying data semantics, since the original layer files record presence/absence of cooperation rather than intensity.

Synthetic 1 (unweighted directed and undirected). Generated using the same SBM specification as the weighted variants but with binary edges (Bernoulli draws on the block probability matrix, no Poisson weight assignment). The unweighted directed and undirected variants are generated as a pair through the same symmetrization procedure as the weighted versions. Provides controlled unweighted networks at scale ($N = 600$).

Synthetic 2 (unweighted directed and undirected). Generated using the same preferential attachment specification as the weighted variants but with binary edges. Provides controlled high-centralization unweighted networks at medium scale ($N = 250$).

A note on thresholded vs. natively-unweighted networks. Thresholding a weighted network produces a structurally different network than one that was unweighted by construction: edge density tends to be higher (any non-zero weight becomes an edge), reciprocity is computed on edge presence rather than weight, and degree distributions reflect edge counts rather than weight sums. The validation interprets unweighted variants as their own networks rather than as representations of the weighted versions, and reports results separately rather than treating them as paired observations of the same underlying structure.

3.4 Corpus Summary

The corpus brings together five empirical networks and two parameterized synthetic families spanning four substantive structural conditions. **Moreno High School** is a friendship network among 70 adolescents collected via standard sociometric instrument, representing the small-but-densely-connected case characteristic of school-based peer studies and providing reciprocity structure with which to test directed-graph measures. **Balikatan 2022** is a counter-insurgency interaction network derived from joint US–Philippines military exercise records ($N = 1,347$, $|E| = 3,146$), representing the large-and-sparse case that arises in operational social-network analysis where ties are recorded across many actors but at low density per actor. **Marvel Universe** is a character co-appearance network ($N = 6,486$, $|E| = 77,227$), representing the large-and-clustered undirected case: it is the corpus’s largest network by a wide margin, with high transitivity (0.79) and a heavy-tailed weight distribution (Gini 0.54), and it complements Balikatan by occupying the opposite topology corner (dense and clustered rather than sparse). **Toledo Crime** is a co-offending network among 77 individuals appearing together in police records, representing the small-and-undirected case typical of administrative-data social networks where tie direction is unavailable or ill-defined. **Scotland Interlock** is a corporate-board interlock projection on 108 directors with ground-truth community structure available from the original bipartite affiliation data, providing a network where the validation framework’s recovered partition can be compared against a known true partition. Each empirical network is included with both its native edge weights and a binary version where weights are stripped; the validation interprets these as separate networks rather than as paired observations. The two synthetic families — a stochastic block model (Synthetic 1) and a preferential attachment generator (Synthetic 2), each emitted in directed and undirected variants — anchor controlled low- and high-centralization endpoints, bracketing the empirical centralization range. Together the corpus covers the four cells of the weighted $\times$ directed design and the full range of low-to-high degree centralization, plus extreme cases at the large-sparse (Balikatan), large-clustered (Marvel), and small-tree-like (Toledo, bicomponent fraction 0.27) corners of the topology space.

The sixteen GraphML files reside in the project’s processed-data directory and have been validated by round-trip serialization tests (write to GraphML, load from GraphML, compare against in-memory representation): all node and edge attributes, weights, and graph-level metadata round-trip cleanly. The synthetic networks additionally satisfy their design specifications: Synthetic 1 has Leiden modularity 0.50 with six recovered communities at $\gamma = 1.0$, matching the six planted blocks; Synthetic 2 has maximum degree 43 and a top-

Network	Source	N	Direction	Weights	$ E $	Gini	Role
Moreno High School	Empirical	70	Directed	Weighted	366	0.24	Small directed dense
Moreno (unweighted)	Threshold	70	Directed	Unweighted	366	0.24	Unweighted directed
Toledo Crime (collapsed)	Empirical	77	Undirected	Weighted	106	0.46	Small undirected real
Toledo (unweighted)	Threshold	77	Undirected	Unweighted	106	0.46	Unweighted undirected
Scotland Interlock (proj.)	Empirical	108	Undirected	Weighted	276	0.48	Comm. truth available
Balikatan 2022	Empirical	1,347	Directed	Weighted	3,146	0.70	Large directed sparse
Marvel Universe	Empirical	6,486	Undirected	Weighted	77,227	0.54	Large undirected clustered
Marvel (unweighted)	Threshold	6,486	Undirected	Unweighted	77,227	0.54	Unweighted undir. at scale
Synthetic 1 (directed)	SBM	600	Directed	Weighted	8,277	0.10	Large directed controlled
Synthetic 1 (undirected)	SBM sym.	600	Undirected	Weighted	4,138	0.14	Large undirected controlled
Synthetic 1 (dir., unwt.)	SBM bin.	600	Directed	Unweighted	8,253	0.10	Unwt. dir. at scale
Synthetic 1 (undir., unwt.)	SBM bin. sym.	600	Undirected	Unweighted	4,228	0.14	Unwt. undir. at scale
Synthetic 2 (directed)	PA	250	Directed	Weighted	497	0.38	Mid directed extreme
Synthetic 2 (undirected)	PA sym.	250	Undirected	Weighted	497	0.38	Mid undirected extreme
Synthetic 2 (dir., unwt.)	PA bin.	250	Directed	Unweighted	497	0.37	Unwt. high-cent. dir.
Synthetic 2 (undir., unwt.)	PA bin. sym.	250	Undirected	Unweighted	497	0.37	Unwt. high-cent. undir.

Table 1: Validation network corpus, 16 networks across the four weighted $\times$ directed cells. Edge counts ($|E|$) and Gini coefficients are computed from the assembled GraphML files. The Gini column reports the coefficient of the unweighted-degree distribution for the synthetic and unweighted-empirical rows, and of the edge-weight distribution for the weighted-empirical rows where noted; < 0.20 indicates low centralization (uniform degree), > 0.35 indicates high centralization (hub-skewed degree). The empirical corpus spans the full range from uniform to highly hub-skewed and adds Marvel at the large-clustered extreme. Synthetic 1 anchors the low-centralization endpoint (Gini 0.10–0.14); Synthetic 2 anchors the high-centralization endpoint (Gini 0.37–0.38). Together the corpus tests the validation framework across a broad swath of the structural-coverage space.

five degree distribution of (43, 38, 35, 18, 13), confirming the hub-spoke structure expected of preferential attachment.

3.5 Phase 0: Preliminary Corpus Characterization

As a preliminary step before the main validation runs, Phase 0 computes descriptive statistics for all networks in the corpus using the measures from [Smith et al. \(2022\)](#): in/out/symmetric degree centrality, closeness, betweenness, Bonacich power, the corresponding centralizations, transitivity, mean inverse distance, component and bicomponent structure, and the ranked-cluster tau statistic. Phase 0 produces an empirical analog of [Smith et al. \(2022\)](#)’s Table 1 for the validation corpus, extended with weighted and triad-census characterizations that the framework’s distinctive capabilities require.

Phase 0 reporting is organized into the opening corpus inventory (Table 1) plus eight descriptive tables. The first four split the centrality-and-topology battery across the 2×2 of directedness $\times$ weighting (Tables 2–5); the next two report the static (unweighted) triad census split by directedness (Tables 6–7); the last two report the weighted layered triad census, again split by directedness (Tables 8–9). This structure supersedes the single binarized table of earlier drafts, which relied on a binarization-invariance footnote to fold weighted and unweighted variants into one column. Separating binary from weighted explicitly is necessary now that the framework computes genuinely weighted path-based measures (closeness, betweenness, and mean inverse distance under tie-strength edge semantics), whose values differ from their binarized counterparts and could not be represented in a binarization-invariant table. The triad census is split by directedness because the directed and undirected cases have different sets of non-zero Davis–Leinhardt classes (sixteen versus four) and dramatically different τ -grid scales, which would render uninformatively if pooled.

Within each of the four measure tables, the empirical networks appear in the leftmost columns and the synthetic networks in the rightmost columns, separated by a vertical rule. This grouping makes the design argument legible at a glance: the empirical networks span a wide range of structural conditions, and the synthetic networks anchor controlled low- and high-centralization endpoints of that range. Reading horizontally across an empirical-then-synthetic row shows whether the empirical variation lies within, outside, or astride the synthetic controls.

Binary versus weighted reporting conventions. The two binary tables (2, 3) compute every measure on the binarized adjacency, matching the convention of Smith et al. (2022)’s Table 1 and ensuring cross-network comparability; the weighted and unweighted variants of a network are binarization-invariant and so each network appears once. The two weighted tables (4, 5) report measures that depend on edge weights and therefore include only the genuinely weighted networks: the unweighted variants would collapse to their binary values (every weight equal to one), so they are omitted rather than shown as redundant or degenerate rows. Two weight-handling conventions are worth making explicit, because they differ across the table families. First, the weight-distribution summaries (min, max, median, IQR, Gini) are computed on the *aggregated directed edge weights* for directed networks and on the *nonzero weights of the symmetrized adjacency’s upper triangle* for undirected networks — the latter matching the graph that the undirected weighted path-based measures actually traverse, where reciprocal arcs are combined by $\max(A, A^\top)$. Second, the weighted path-based measures symmetrize undirected graphs by $\max(A, A^\top)$, whereas the weighted triad census symmetrizes by sum $(W + W^\top)$ before thresholding, following *s*-core semantics; the two table families therefore treat undirected edge weights differently by design, and values should not be expected to align across them.

Measure	Empirical			Synthetic	
	Marvel	Scot	Toledo	SBM-u	PA-u
<i>Centrality, mean (SD)</i>					
Degree	11.91 (31.71)	2.56 (2.31)	1.38 (1.70)	7.05 (1.75)	1.99 (2.28)
Closeness	0.19 (0.12)	0.26 (0.14)	0.34 (0.07)	0.38 (0.02)	0.32 (0.05)
Betweenness	0.00 (0.00)	0.01 (0.02)	0.03 (0.10)	0.00 (0.00)	0.01 (0.03)
Bonacich Power	0.03 (0.07)	0.19 (0.19)	0.14 (0.14)	0.51 (0.13)	0.10 (0.11)
<i>Centralization (SD of node scores)</i>					
Degree	31.71	2.31	1.70	1.75	2.28
Closeness	0.13	0.14	0.07	0.02	0.05
Betweenness	0.00	0.02	0.10	0.00	0.03
Bonacich Power	0.07	0.19	0.14	0.13	0.11
<i>Topology</i>					
Component (prop.)	0.70	0.80	1.00	1.00	1.00
Bicomponent (prop.)	0.65	0.73	0.27	1.00	1.00
Mean Inv. Distance	0.05	0.06	0.05	0.13	0.02
Transitivity	0.79	0.46	0.42	0.06	0.10

Table 2: Phase 0 Table 1 — undirected, binary. Centrality and topology measures computed on the binarized adjacency for the undirected networks, empirical (left) and synthetic (right). Each column is a single network; weighted and unweighted variants are binarization-invariant and so appear once. Centrality cells show mean (SD) of node-level scores; centralization is the network-level standard deviation of those scores, following Smith et al. (2022). Component and bicomponent sizes are reported as proportions of the node set in the largest (bi)component. Marvel anchors the large-clustered extreme (transitivity 0.79, $N = 6,486$); Synthetic 1 (SBM-u) and Synthetic 2 (PA-u) anchor the low- and high-centralization synthetic endpoints.

Measure	<i>Empirical</i>		<i>Synthetic</i>	
	Moreno	Bali	SBM-d	PA-d
<i>Centrality, mean (SD)</i>				
In-Degree	5.23 (3.64)	2.34 (2.70)	13.76 (3.52)	1.99 (4.65)
Out-Degree	5.23 (2.04)	2.34 (18.55)	13.76 (3.36)	1.99 (0.14)
Symmetric Degree	10.46 (4.56)	4.64 (19.11)	27.51 (4.98)	3.98 (4.55)
Closeness	0.45 (0.06)	0.23 (0.12)	0.45 (0.02)	0.32 (0.05)
Betweenness	0.04 (0.05)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Bonacich Power	0.39 (0.19)	0.02 (0.04)	0.58 (0.11)	0.10 (0.11)
<i>Centralization (SD of node scores)</i>				
In-Degree	3.64	2.70	3.52	4.65
Out-Degree	2.04	18.55	3.36	0.14
Symmetric Degree	4.56	19.11	4.98	4.55
Closeness	0.06	0.12	0.02	0.05
Betweenness	0.05	0.00	0.00	0.00
Bonacich Power	0.19	0.04	0.11	0.11
<i>Topology</i>				
Component (prop.)	1.00	0.81	1.00	1.00
Bicomponent (prop.)	1.00	0.49	1.00	1.00
Mean Inv. Distance	0.45	0.23	0.45	0.32
Transitivity	0.40	0.23	0.05	0.06
Tau_{RC}	8.73	2535.93	−6.82	63.03

Table 3: Phase 0 Table 2 — directed, binary. Measures computed on the binarized adjacency for the directed networks, empirical (left) and synthetic (right). Centrality cells show mean (SD); centralization is the network-level SD of node scores. Component and bicomponent sizes are proportions of the node set. Mean inverse distance is the mean reciprocal geodesic distance over ordered pairs. Tau_{RC} is the standardized deviation of the observed ranked-cluster-weighted triad count from its U|MAN null expectation, by Monte Carlo with 500 samples; its magnitude scales with $\sqrt{N}$ for very-sparse networks, explaining the large Balikpapan value. Tau_{RC} is defined only for directed graphs (it requires the mutual–asymmetric dyad distinction) and so appears in this table but not Table 2.

Measure	<i>Empirical</i>			<i>Synthetic</i>	
	Marvel	Scot	Toledo	SBM-u	PA-u
<i>Weight distribution</i>					
Minimum	2.00	1.00	1.00	1.00	1.00
Maximum	744.00	8.00	5.00	9.00	6.00
Median	3.00	1.00	2.00	3.00	2.00
IQR	3.00	0.00	2.00	2.00	2.00
Gini	0.54	0.24	0.26	0.26	0.26
<i>Centrality, mean (SD)</i>					
Strength	73.81 (333.28)	3.45 (3.63)	2.90 (4.14)	20.65 (5.88)	3.92 (4.24)
Wtd. Closeness	1.28 (1.18)	0.30 (0.17)	0.67 (0.24)	1.30 (0.09)	0.60 (0.12)
Wtd. Betweenness	0.00 (0.00)	0.01 (0.02)	0.03 (0.10)	0.00 (0.00)	0.01 (0.03)
Bonacich Power	0.03 (0.07)	0.19 (0.19)	0.14 (0.14)	0.56 (0.14)	0.11 (0.11)
<i>Centralization (SD of node scores)</i>					
Strength	333.28	3.63	4.14	5.88	4.24
Wtd. Closeness	1.18	0.17	0.24	0.09	0.12
Wtd. Betweenness	0.00	0.03	0.10	0.00	0.04
Bonacich Power	0.07	0.19	0.14	0.14	0.11
<i>Topology</i>					
Wtd. Mean Inv. Distance	0.28	0.07	0.09	0.40	0.04

Table 4: Phase 0 Table 3 — undirected, weighted. Weighted measures for the genuinely weighted undirected networks; unweighted variants are omitted because their weighted measures collapse to the binary values of Table 2. Weighted closeness, betweenness, and mean inverse distance use tie-strength edge semantics (Dijkstra on $1/w$, so stronger ties are shorter). Strength is weight-summed degree. The weight-distribution block (min, max, median, IQR, Gini) is computed on the nonzero weights of the symmetrized adjacency’s upper triangle ($\max(A, A^\top)$), the graph the undirected weighted measures traverse. Marvel’s dispersed weights (max 744, Gini 0.54) drive its high mean strength and weighted closeness relative to the binary case.

Measure	<i>Empirical</i>		<i>Synthetic</i>	
	Moreno	Bali	SBM-d	PA-d
<i>Weight distribution</i>				
Minimum	1.00	1.00	1.00	1.00
Maximum	2.00	31.00	10.00	6.00
Median	1.00	1.00	3.00	2.00
IQR	1.00	0.00	2.00	2.00
Gini	0.17	0.27	0.26	0.26
<i>Centrality, mean (SD)</i>				
In-Strength	7.23 (5.38)	3.35 (6.75)	41.53 (11.78)	3.92 (8.60)
Out-Strength	7.23 (2.65)	3.35 (30.26)	41.53 (12.04)	3.92 (1.43)
Total Strength	14.46 (6.93)	6.62 (32.79)	83.07 (17.00)	7.83 (8.48)
Wtd. Closeness	0.59 (0.08)	0.35 (0.20)	1.69 (0.09)	0.60 (0.12)
Wtd. Betweenness	0.04 (0.05)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Bonacich Power	0.39 (0.19)	0.02 (0.04)	0.60 (0.11)	0.11 (0.11)
<i>Centralization (SD of node scores)</i>				
In-Strength	5.38	6.75	11.78	8.60
Out-Strength	2.65	30.26	12.04	1.43
Total Strength	6.93	32.80	17.00	8.48
Wtd. Closeness	0.08	0.20	0.09	0.12
Wtd. Betweenness	0.05	0.00	0.00	0.00
Bonacich Power	0.19	0.04	0.11	0.11
<i>Topology</i>				
Wtd. Mean Inv. Distance	0.59	0.35	1.69	0.60

Table 5: Phase 0 Table 4 — directed, weighted. Weighted measures for the genuinely weighted directed networks; unweighted variants are omitted (their weighted measures collapse to Table 3). Weighted path-based measures use tie-strength semantics (Dijkstra on $1/w$). In-, out-, and total strength are the weight-summed in-, out-, and total degree. The weight-distribution block is computed on the aggregated directed edge weights. Moreno’s narrow weight range (1–2, recording single vs. double nominations) yields the lowest weight Gini; Balikpapan’s interaction counts are the most dispersed in the out-direction (out-strength centralization 30.26), reflecting a few very high-volume broadcasting accounts.

DL Class	<i>Empirical</i>		<i>Synthetic</i>	
	Moreno	Bali	SBM-d	PA-d
003	37,827	4.03×10^8	31,233,805	2,453,719
012	9,822	3,602,267	4,204,617	115,292
102	5,175	38,837	158,702	0
021D	170	225,413	47,815	196
021U	311	3,865	49,106	2,886
021C	323	14,258	101,315	849
111D	430	127	7,329	0
111U	214	3,334	6,997	0
030T	82	3,400	6,648	58
030C	5	2	2,369	0
201	67	15	227	0
120D	89	65	277	0
120U	41	669	247	0
120C	36	3	684	0
210	92	9	61	0
300	56	5	1	0

Table 6: Phase 0 Table 5a — static triad census (directed networks), Davis–Leinhardt class counts. A single Batagelj–Mrvar census on the binarized graph; each cell is the raw count of triples in that class. All sixteen DL classes appear; counts span several orders of magnitude and are reported with thousand separators (scientific notation for cells $\geq 10^8$). The closed-triangle count (class 300) matches the unweighted triangle count $T_{\max}$ used by the layered census in Table 8.

DL Class	<i>Empirical</i>			<i>Synthetic</i>	
	Marvel	Scot	Toledo	SBM-u	PA-u
003	4.50×10^{10}	176,906	65,795	33,349,744	2,453,752
102	4.74×10^8	25,513	6,789	2,413,690	115,292
201	11,718,673	1,468	537	55,644	3,904
300	1,028,453	269	29	1,122	52

Table 7: Phase 0 Table 5b — static triad census (undirected networks), Davis–Leinhardt class counts. Only the four undirected isomorphism classes $\{003, 102, 201, 300\}$ are non-zero on undirected graphs; the twelve asymmetric DL classes are zero by construction and omitted. Counts are computed by the dyad-driven Batagelj–Mrvar kernel applied to the symmetrized adjacency, validated against an external Pajek reference on Balikpapan and against the reference triple-loop kernel on Scotland. Marvel’s $300 = 1,028,453$ closed triangles match the unweighted triangle count $T_{\max}$ used by the layered census in Table 9.

Quantity	<i>Empirical</i>		<i>Synthetic</i>	
	Moreno	Bali	SBM-d	PA-d
<i>Grid metadata</i>				
$T_{\max}$	307	3,015	7,001	42
L	10	8	8	16
$\tau_{\min}$	1.00	1.26	2.15	1.00
$\tau_{\max}$	2.00	4.97	4.64	5.00
Method	empirical	analytic	analytic	empirical
Wall-clock (s)	0.07	0.24	0.06	0.02
<i>Per-class AUMC and peak τ</i>				
AUMC ₀₀₃	0.259	0.596	0.315	0.687
peak τ_{003}	1.08	4.08	4.16	5.00
AUMC ₀₁₂	0.0271	5.63×10^{-4}	0.0179	0.0117
peak τ_{012}	1.00	1.26	2.15	1.00
AUMC ₁₀₂	0.0124	1.05×10^{-5}	3.22×10^{-4}	0
peak τ_{102}	1.00	1.26	2.15	1.00
AUMC _{021D}	9.35×10^{-5}	9.03×10^{-6}	1.05×10^{-4}	1.16×10^{-5}
peak τ_{021D}	1.00	1.26	2.15	1.00
AUMC _{021U}	3.70×10^{-4}	3.71×10^{-7}	1.06×10^{-4}	1.43×10^{-4}
peak τ_{021U}	1.00	1.26	2.15	1.00
AUMC _{021C}	3.84×10^{-4}	1.44×10^{-6}	2.23×10^{-4}	4.50×10^{-5}
peak τ_{021C}	1.00	1.26	2.15	1.00
AUMC _{111D}	5.05×10^{-4}	4.37×10^{-8}	8.52×10^{-6}	0
peak τ_{111D}	1.00	1.26	2.15	1.00
AUMC _{111U}	1.69×10^{-4}	1.93×10^{-7}	8.46×10^{-6}	0
peak τ_{111U}	1.00	1.26	2.15	1.00
AUMC _{030T}	6.66×10^{-5}	2.84×10^{-7}	7.72×10^{-6}	1.58×10^{-6}
peak τ_{030T}	1.00	1.26	2.15	1.00
AUMC _{030C}	1.53×10^{-6}	0	2.79×10^{-6}	0
peak τ_{030C}	1.00	1.26	2.15	1.00
AUMC ₂₀₁	7.24×10^{-5}	2.94×10^{-9}	1.75×10^{-7}	0
peak τ_{201}	1.00	1.26	2.15	1.00
AUMC _{120D}	1.52×10^{-4}	2.62×10^{-9}	1.84×10^{-7}	0
peak τ_{120D}	1.00	1.26	2.15	1.00
AUMC _{120U}	5.93×10^{-5}	2.11×10^{-8}	1.26×10^{-7}	0
peak τ_{120U}	1.00	1.26	2.15	1.00
AUMC _{120C}	2.66×10^{-5}	1.47×10^{-9}	5.20×10^{-7}	0
peak τ_{120C}	1.00	1.26	2.15	1.00
AUMC ₂₁₀	1.16×10^{-4}	9.44×10^{-10}	3.26×10^{-8}	0
peak τ_{210}	1.00	1.26	2.15	1.00
AUMC ₃₀₀	4.83×10^{-5}	0	0	0
peak τ_{300}	1.00	1.26	2.15	1.00

Table 8: Phase 0 Table 6a — weighted layered triad census (directed networks). For each network the layered census thresholds the weighted adjacency across a log-spaced grid of L thresholds $\tau \in [\tau_{\min}, \tau_{\max}]$ chosen by `recommend_L`; per-class AUMC values are areas under the density curves on $\log\text{-}\tau$, and peak τ is the threshold at which each class’s density peaks. AUMC values span several orders of magnitude (the empty-triad class 003 dominates while the dense motifs 201 and 300 are rare) and are reported in scientific notation; peak τ is reported in the native τ units. The method dispatch (analytic vs. empirical) is set by `recommend_L` based on $T_{\max}$ and the weight-distribution span. Unweighted variants of these networks are omitted because their weighted measures collapse to the static counts of Table 6.

Quantity	<i>Empirical</i>			<i>Synthetic</i>	
	Marvel	Scot	Toledo	SBM-u	PA-u
<i>Grid metadata</i>					
$T_{\max}$	1,028,453	269	29	1,024	42
L	14	16	10	16	16
$\tau_{\min}$	4.40	1.00	1.00	1.00	1.00
$\tau_{\max}$	227.75	8.00	4.00	7.00	5.00
Method	analytic	empirical	empirical	empirical	empirical
Wall-clock (s)	1.59	0.01	0.00	0.14	0.02
<i>Per-class AUMC and peak τ</i>					
AUMC ₀₀₃	1.71	0.889	0.572	0.816	0.687
peak τ_{003}	227.75	5.28	3.43	6.15	4.03
AUMC ₁₀₂	1.15×10^{-3}	0.0135	0.028	0.0286	0.0117
peak τ_{102}	4.40	1.00	1.00	1.00	1.00
AUMC ₂₀₁	9.19×10^{-6}	2.81×10^{-4}	1.46×10^{-3}	4.71×10^{-4}	1.99×10^{-4}
peak τ_{201}	4.40	1.00	1.00	1.00	1.00
AUMC ₃₀₀	1.31×10^{-6}	6.71×10^{-5}	1.19×10^{-4}	7.00×10^{-6}	1.58×10^{-6}
peak τ_{300}	4.40	1.00	1.00	1.00	1.00

Table 9: Phase 0 Table 6b — weighted layered triad census (undirected networks). Only the four undirected isomorphism classes $\{003, 102, 201, 300\}$ are non-zero on undirected graphs; the twelve asymmetric DL classes are zero across the τ grid by construction and omitted. Weights are symmetrized by sum $(W + W^T)$ before thresholding, following s -core semantics — a different convention from the $\max(A, A^T)$ symmetrization used by the weighted path-based measures in Tables 4–5. AUMC values are reported in scientific notation, peak τ in native units. Marvel’s $T_{\max} = 1,028,453$ and analytic τ grid span the widest range in the corpus.

4 Degradation Strategy

For each network in the corpus the validation treats the network as the true network T , induces missingness to produce a degraded observation G_{obs} , and hands G_{obs} to the framework and the baselines for reconstruction against the known truth. The principle behind the degradation is that the stimulus should fix exactly the quantities the framework’s prior encodes — the amount of missingness and its correlation with centrality — while leaving free everything the framework is meant to recover or exploit. That principle determines which conditions the sampler enforces and which it merely records.

4.1 The sampler and its two arms

Missingness is induced by a propensity field shared with the reconstruction engine. Given the observed centrality (in-degree for directed networks, degree for undirected), a target ρ , and a target missingness level, the field solver — the same `_solve_propensity_field` the reconstruction setup uses — returns a per-node selection tendency by analytically inverting the Kendall τ_b relationship between the missing-node indicator and centrality. Degradation is, by construction, the inverse of reconstruction under one field.

The sampler runs in one of two arms, selected by `node_loss`:

- **Edge arm** (:emergent). Edge weight is removed in proportion to current weight, tilted by the field, following Bellutta (2026); a node is lost only when all of its incident weight reaches zero. The missingness proportion and ρ are *outcomes* of the weight removal, not targets. Because orphaning a node costs its entire incident weight, hubs survive and the lost set skews peripheral, so the arm reaches mostly negative ρ .

- **Node arm** (:targeted). The missing set is topped up to the target proportion by field-weighted selection, so the proportion and ρ are targeted directly, and the arm reaches both signs of ρ .

A missing node is then materialized by `_materialize_missing_nodes`, which decides per node between two fates: a node that still has surviving in-edges (some observed actor still names it) is retained as a partially observed *nominated non-respondent*, with its outgoing ties removed; a node with no surviving nominator is removed entirely. The two arms fall out of this naturally — edge-arm losses, having shed all incident weight, are full removals, while node-arm targets that others still name are retained as nominated non-respondents. Removal is not unconditional.

4.2 What the degradation conditions on

The degradation ends in a single acceptance gate whose behavior depends on the arm.

In the **node arm** (:targeted), where the proportion and ρ are targets, the gate enforces all three priors:

- **Missingness proportion** — realized $|\text{missing}|/N$ within $1/N$ of the target.
- **Centrality–missingness correlation** — realized Kendall τ_b within `rho_tol` of the target.
- **E/I-by-degree-rank distortion** — the occupancy-weighted per-rank-bin TVD between the surviving sub-graph’s E/I-by-rank profile and the true network’s, within `ei_tvd_tol` (default 0.25). A removal that strips brokers and bends the high-rank E/I relationship away from the truth is rejected; a benign removal that leaves the curve intact passes. This prior auto-passes where it cannot be assessed — a single community, or no survivor rank bin clearing the minimum count — which is the high-missingness regime where community structure ceases to be meaningful.

In the **edge arm** (:emergent), where the proportion and ρ are emergent outcomes of weight removal rather than targets, the gate enforces nothing: all three quantities are recorded but none gates. The E/I distortion in particular is deliberately record-only here, because gating it would admit only distortion-preserving draws and rig the stimulus.

Mechanically, the gate runs inside a bounded retry loop. A draw that misses an enforced prior is re-rolled under a fresh per-attempt seed up to `max_retries`; the accepted (or closest) draw is recorded with its `gate_status`, the field-solve `field_status`, the realized proportion, the realized ρ (Kendall τ_b), the recorded E/I distortion, and the edge-degradation diagnostics (W_{true} , W_{removed} , the zeroed-edge and organic-loss counts, and β).

4.3 The Degraded-Network Stimulus

The sampler is run ahead of time to build a fixed corpus of degraded networks — the *stimulus* — that the framework and the baselines are evaluated against, so every method sees the same inputs.

ρ is withheld truth. For each degraded sample, ρ (the Kendall τ_b of the missing-node indicator against centrality) is the realized truth of that sample, and it is withheld from reconstruction: recorded only to score, never supplied, because an analyst cannot measure it in practice — it lives in the data they did not collect. Reconstruction instead receives an *assumed* ρ , and the experiment compares a correct assumption against the cost of getting it wrong in each direction: assuming missingness is random when it is signed, positive when it is negative, and negative when it is positive. Assumed and true ρ are the same binary- τ_b -against-centrality quantity, so the two line up cleanly — a +0.35 assumption is scored against a +0.35 truth.

Corpus and grid. The same eighteen networks are degraded under both arms. They fill a 2×2 of weight heterogeneity $\times$ degree heterogeneity — hub-periphery (Marvel, Balikpapan, Toledo, and the two preferential-attachment synthetics) against homogeneous (Scotland, the two SBM synthetics, and Moreno) — each present in unweighted and weighted form. Missingness is swept over six levels, 1, 10, 20, 30, 40, 50% (of total weight for the edge arm, of nodes for the node arm). The ρ targets are $\{-0.35, -0.15, 0\}$ for the edge arm (negative-only, reflecting its reach) and $\{-0.35, -0.15, 0, +0.15, +0.35\}$ for the node arm (symmetric). Generation is keep-all-then-select: a pool of two hundred replicates per (network, level, nominal- ρ) cell is drawn over a fine nominal grid, and point conditions are selected post-hoc at each target ρ from the draws whose realized ρ falls within ± 0.03 (snapping to the nearest achievable τ_b on small or discrete networks), seeded for reproducibility. One reconstruction guess per condition then stands in for every sample in it.

Reachable ρ . The arms’ reach is asymmetric, and in opposite directions. In the production pool the edge arm reaches the full $\{-0.35, -0.15, 0\}$ set on fourteen of the eighteen networks; on Marvel (both variants) and the unweighted directed SBM it reaches only $\{-0.15, 0\}$, on the weighted directed SBM only $\{0\}$, and positive ρ nowhere — its weight-cost rule cannot orphan hubs. The node arm reaches the full symmetric set on all eighteen once $\pi_{\text{node}} \geq 0.20$; at $\pi_{\text{node}} = 0.10$ it reaches $+0.35$ everywhere but falls just short of -0.35 on the preferential-attachment and Toledo variants (about -0.30 to -0.32). Both patterns trace to the heavy tail, where a few rank-distinct hubs sit over a large, degree-tied periphery. For the node arm, pushing ρ positive means dropping the hubs: because they are rank-distinct, a handful of removals already yields a clean τ_b , so positive targets are reached at the lowest missingness. Pushing ρ negative means dropping the periphery, where the degree ties attenuate τ_b , so a strong negative value needs more removals — which is why -0.35 clears only at $\pi_{\text{node}} \geq 0.20$. The edge arm reads the same structure through a rule that cannot orphan hubs at all, so it sheds only that periphery and reaches negative ρ alone: the same asymmetry, pointed the other way.

Giant-component collapse. At high node-missingness a real fraction of the sparse hub-periphery draws fragment: the dense homogeneous networks never collapse, but Toledo loses up to forty percent of its draws by $\pi_{\text{node}} = 0.50$, with the other sparse networks in between. This dovetails with the E/I design point above — it is exactly where community structure begins to become inapplicable — and whether to exclude collapsed draws or record collapse as an outcome is left open here.

5 Degradation Tests

The degradation half of the pipeline is verified against the current implementation by three groups of tests: a sampler-correctness suite (`run_phase1_sampler_tests`) on small constructed fixtures and corpus networks, an orchestrator smoke run over the two-dial grid, and a set of emergent-arm tests (`run_emergent_checks`) that pin down the edge arm’s distinctive behavior. The sampler tests are fast and seed-deterministic; the orchestrator and emergent runs exercise `build_degeneration_corpus` end to end on a small grid.

5.1 Sampler correctness

The core suite checks that the propensity-field sampler hits its targets, honors its ceilings, and fails loudly where it must. The fixtures are Scotland and Moreno from the corpus plus two constructed extremes — a 50-node star (one hub, all else degree one) and a 50-node 4-regular ring (constant degree).

- **ρ convergence** (Scotland). Across replicates at a feasible target, the mean realized Kendall τ_b matches the target within tolerance — the field hits the requested ρ on average.

- **Rate-bounded ceiling** (Scotland). A target above the network’s practical ceiling but below the rate-bounded absolute ceiling $\sqrt{2p(1-p)}$ is detected: the sampler reports the achieved ceiling value through its field-solve status rather than the unattained nominal target.
- **Exact proportion** (Scotland and Moreno). In the targeted arm, every replicate’s realized count equals $\text{round}(\text{rate} \cdot N)$ exactly, asserted per replicate rather than on the mean.
- **Achievable- ρ ceiling** (star). On a single-hub star the centrality distribution cannot support an extreme target, and the ceiling path engages rather than the sampler silently reporting the target.
- **Seed reproducibility** (Moreno). Identical $(\text{rate}, \rho, \text{seed})$ produce bit-identical records — dropped set, realized values, and status.
- **MCAR baseline** (Scotland). At $\rho = 0$ the mean realized ρ is near zero, confirming uniform selection.
- **Constant centrality** (4-regular ring). A centrality vector with no variance admits no correlation; the sampler handles this explicitly rather than fabricating a realized ρ .
- **Degeneracy flagging**. An adversarial high-rate cell trips the topological-degeneracy flags (`gc_collapse`, `too_small`, `no_edges`), confirming the detector fires where the degraded network has genuinely fragmented.
- **Materialization composition**. `_materialize_missing_nodes` retains a missing node that still has surviving in-edges as a partially observed nominated non-respondent and fully removes one with no surviving nominator — the auto-nomination behavior the strategy section relies on.
- **Weight-stage contract**. The edge-degradation weight accounting holds across seeds: removed and surviving weight reconcile against the recorded W_{true} and W_{removed} .
- **Prior-3 survivor profile**. The E/I-by-degree-rank distortion is recomputed on the thinned subgraph against the precomputed true profile, and gates under `:targeted` at `ei_tvd_tol = 0.25` — a broker-stripping removal is rejected, a benign one passes.

5.2 Orchestrator smoke run

A pre-launch smoke run drives `build_degeneration_corpus` over a small two-dial $(\rho \times \pi_{\text{node}})$ grid — explicitly with no mechanism axis, since materialization is a single auto-nomination step rather than two parallel mechanisms. The run confirms that the orchestrator produces well-formed records for every cell, that the recorded fields (nominal targets, realized proportion and ρ , gate and field statuses, degeneracy flags) are populated as expected, and that degraded samples can be materialized back from the stored records for inspection.

5.3 Emergent-arm behavior

The final group isolates the edge arm’s defining properties, which the targeted arm does not share.

- **Weight drop without node loss**. At a moderate π_{edge} , weight is removed but no node sheds all of its incident weight, so the emergent missing set stays empty — a node is lost only when its incident weight reaches zero.

- **π_{node} is a tilt, not a target.** In :emergent the node-missingness parameter only tilts the field; there is no top-up to a target proportion, and the realized proportion is an outcome.
- **Shared edge stream.** The :emergent and :targeted arms consume the same edge-degradation stream; the targeted top-up is the only difference between them.
- **Gated under targeted, recorded under emergent.** The same broker-distorting removal is rejected by the prior-3 gate under :targeted but accepted and recorded under :emergent — the arm asymmetry the Degradation Strategy section describes, verified directly.

6 Reconstruction Strategy

For each degraded sample in the stimulus the framework is handed the observed network G_{obs} and an *assumed* ρ , and returns a posterior sample of reconstructed networks from which credible intervals on the requested measures are formed. This section specifies the reconstruction pipeline, the assumed-prior design the validation turns on, and how the runs are scoped across the corpus. The reconstruction algorithm itself is specified in the companion document; what follows is how it is exercised in validation.

6.1 The reconstruction pipeline

Reconstruction runs in two phases, orchestrated by `build_reconstruction_corpus`. *Setup* (`compute_setup`) reads the observed network once: it computes observed centrality (in-degree for directed networks, degree for undirected), detects community structure (CHAMP, or supplied labels) and the E/I index, bins the observed nodes into K degree-rank bins by J E/I bins, estimates the Laplace-smoothed rank-rank connectivity matrix P and, for directed networks, the rank-conditional reciprocity matrix R , and solves the propensity field β that realizes the assumed ρ under the analytic Kendall τ_b relation. *Replication* (`generate_replicate`) then draws B augmented networks from that setup: each draw assigns bins from the setup distribution q , adds the missing nodes, and places their ties through Stage 0.5 (rank-conditional reciprocity imputation for nominated non-respondents, directed networks only), Stage 1 (synthetic nodes whose tie counts and partners are drawn against P), and Stage 2 (diffuse edge-weight reconstruction, for weighted networks). Each draw is admitted by a three-prior acceptance gate on the realized proportion, edge weight, and ρ , regenerating on a miss and falling back to the closest draw if the attempt budget is exhausted (the E/I structure enters by construction, through the conditional placement of added nodes, rather than through this gate). The requested measures are computed on each admitted replicate, and the B values form the posterior from which a credible interval is read.

6.2 The assumed prior

The framework never sees the true ρ ; it sees only the analyst’s assumed value, and the validation is built around the gap between them. For each degraded sample the framework is run under three assumed priors:

- **Correct** — the assumed ρ equals the sample’s withheld true ρ . This is the framework’s ceiling.
- **MCAR** — the assumed $\rho = 0$: the analyst with no domain knowledge.
- **Opposite-sign** — the assumed ρ has the true magnitude but the wrong sign: the analyst fundamentally mistaken about the missingness.

Reading coverage and width across the three exposes the cost of a wrong assumption in each direction. When the assumed ρ exceeds what is structurally achievable on G_{obs} at its K and node count, the framework clamps

it to the feasible envelope and conditions the sample on the clamped value, recording nominal, conditioned, and realized ρ ; the interval is then interpreted against the prior actually enforced, not the one requested.

6.3 Replication

For each (degraded sample, assumed prior) the framework draws $B = 1,000$ posterior replicates, and the posterior of each measure over those replicates yields one credible interval. Coverage and width for a condition are aggregated over its point-selected degraded draws — each draw contributing one interval — and can be reported against either the nominal grid or the realized (rate, ρ) of each draw, the latter the more honest summary where the achievable- ρ ceiling keeps the nominal target out of reach. The reconstruction runs share the master-seed discipline of the rest of the pipeline, so any single (sample, prior, replicate) is independently reproducible.

6.4 Scope across the corpus

The three analyses are scoped by cost — broad and cheap first, narrow and expensive last; their metrics are defined in the Tracks section.

- **Network-similarity recovery — all eighteen networks.** At the MCAR condition ($\rho = 0$), where every network and missingness level is in scope, each reconstructed replicate’s similarity to the truth is compared against the degraded network’s, isolating whether reconstruction moves G_{obs} toward T at all.
- **ρ -bias — all eighteen networks, appropriate measures.** For the measures where bias is well-defined, the deviation of the off-truth (MCAR and opposite-sign) reconstructions from the truth-conditioned one quantifies the cost of a misspecified prior.
- **Credible-interval calibration — flagged networks.** The full coverage-and-width analysis is the expensive pass, reserved for the networks the broader two passes flag as warranting a closer look.

7 Reconstruction Tests

The reconstruction half of the pipeline is verified by a thirty-eight-test suite in five tiers, run against the current implementation. The first two tiers hold the setup phase (`compute_setup`) accountable — deterministic unit tests on each helper, then a calibration tier checking the three-prior contract on corpus networks — and the next two mirror that structure for replicate generation (`generate_replicate`). The final tier exercises the acceptance gate, the materialization round-trip, and the corpus orchestrator. The unit tiers are fast and fully deterministic; the calibration tiers run on real networks, reproducible from seeds, with Marvel the long pole at roughly thirty minutes of wall-clock for its CHAMP community detection.

7.1 Setup unit tests (Tests 1–18)

Each test exercises one helper of `compute_setup` on a tractable fixture — a small ring, a directed star, a small SBM with known communities — and asserts the output against a hand-derivable expectation, with every branch (directed vs. undirected, weighted vs. unweighted, the binning-fallback modes, and the `:converged/``:ceiling_hit/``:failed_other` states of the  $\beta$ -solver) getting its own test. The helpers covered are `_compute_observed_centrality` (in-degree for directed, degree for undirected), `_compute_ei_values` (all-internal, all-external, and mixed partitions), `_detect_community_structure` (its `:single_community`, `:too_few_nodes`, and `:insufficient_ei_variance` fallbacks, plus the healthy `:two_dimensional` path), `_bin_observed_nodes` (equal-rank K degree bins, and the $J = 3$ semantic E/I thresholds at ± 0.33),

`_compute_p_matrix` (Laplace smoothing on known counts, and exclusion of partially observed nodes), `_compute_r_matrix` (on a fully reciprocated directed graph), `_determine_n_add` (the node-count arithmetic across edge cases), `_realized_rho_for_beta` (its analytic properties), and `_solve_bin_distribution` (bisection convergence). Test 18 is the end-to-end smoke test on `compute_setup` itself: it runs the full setup on a small fixture and checks that the returned `SamplerSetup` is internally consistent — q sums to one, N_{add} matches the π_{node} arithmetic, the β -status is `:converged` on a feasible target, and each populated row of `ei_conditional` sums to one.

7.2 Setup behavior calibration (Tests 19–21)

The calibration tier holds `compute_setup` to the three-prior contract on real networks: Moreno (directed, $N = 70$), Scotland at $K = 4$ and again at $K = 10$ (the same network at a higher bin count, to exercise the K -dependent ceiling), and Marvel (undirected, $N = 6,486$, heavy-tailed). Each network is run at three cells — ($\rho = 0$, rate = 0.10), ($\rho = +0.5$, rate = 0.10), and ($\rho = -0.5$, rate = 0.10) — and the three priors are checked: Prior 1 (proportion missing) as the arithmetic identity $N_{\text{add}}/(N_{\text{obs}} + N_{\text{add}}) \approx \pi_{\text{node}}$; Prior 2 (centrality–missingness correlation) by re-evaluating `_realized_rho_for_beta` at the solved β against the requested ρ , or, where the target exceeds the achievable range for that K and N_{add} , by confirming a `:ceiling_hit` status with β at the bound; and Prior 3 (E/I conditional on degree bin) by checking that each `ei_conditional` row is stochastic and matches the empirical conditional in G_{obs} . One design point is worth stating: each cell first runs the degradation sampler to produce a realistic (realized ρ , realized rate) pair, which is then fed to setup as its target — so the test holds the reconstruction contract accountable on realistic inputs without conflating it with the degradation mechanism. Realized values are reported alongside the nominal targets, and `:ceiling_hit` counts are surfaced as diagnostics so ceiling-pulled cells stay visible. Test 21 (Marvel) is the stress case.

7.3 Replicate unit tests (Tests 22–28)

This tier mirrors the setup unit tests for the replicate-generation helpers. `_draw_bin_assignments` must produce a histogram matching q within tolerance over 10,000 draws; `_build_augmented_nodes` must assign added-node identifiers $N + 1, \dots, N + N_{\text{add}}$ and tag each row with the correct `:node_type`; the stage helpers (`_stage_0_5_directed`, `_stage_0_5_undirected`, `_stage_1_directed`, `_stage_1_undirected`) must respect their structural invariants — no self-loops, no edges introduced between two observed nodes, and `src < dst` on undirected output; and `_stage_2!` must satisfy the weight-conservation identity, $\sum \text{weight} = W_{\text{aug}} + W_{\text{add}}$, after the call. Test 28 carries tighter arithmetic assertions than the others because Stage 2 has a closed-form expected output once π_{edge} and W_{aug} are fixed.

7.4 Replicate integration and calibration (Tests 29–34)

Tests 29–30 run `generate_replicate` end-to-end on small synthetic fixtures ($N = 30$, one directed and one undirected) across $R = 20$ seeds, checking composition: every replicate has $N + N_{\text{add}}$ augmented nodes with exactly N_{add} tagged `:added`, no self-loops, no respondent–respondent edges introduced by Stage 1, bit-identical output under a repeated seed, and a mean realized τ_b across the twenty seeds that matches `setup_rho`. Tests 31–34 are the replicate-generation analogs of the setup calibration — the same four networks, the same three cells, $R = 20$ — but now Prior 2 is measured on each replicate’s *realized* Kendall τ_b , computed from the augmented nodes’ added-indicator against their degree bin on the actual augmented network, rather than on bin-distribution samples in isolation. This catches composition faults the helper tests cannot: a mis-sampled q , a mis-built realized- ρ mask, or a stage that drops or duplicates added rows.

7.5 Corpus and gate coverage (Tests 35–38)

The final tier covers the pieces that turn replicates into a corpus. Test 35 unit-checks `_passes_three_prior_gate`, the acceptance gate. Test 36 verifies the materialization round-trip — `_extract_reconstruction_delta` followed by `materialize_reconstruction` reproduces the augmented network exactly — so that storing deltas rather than whole networks is lossless. Test 37 is a regression guard on `_aggregate_duplicate_dyads`, confirming it preserves the node-identifier type. Test 38 exercises `build_reconstruction_corpus` end-to-end: gate acceptance, ρ conditioning (clamping an infeasible request to the feasible envelope), and reproducibility from a master seed.

8 Comparison Baselines

The framework is compared against five baselines representing the realistic alternatives a practitioner would consider.

Listwise deletion (floor baseline). The reduced network G_{obs} is used directly without imputation. Measures are computed on G_{obs} as-is. This represents the no-method case and establishes the magnitude of the missing-data problem the framework is addressing.

Asymmetric imputation (directed networks only). Following [Smith et al. \(2022\)](#): when an observed forward edge points to a removed node, no reverse edge is added. This privileges observed structure at the cost of systematic asymmetry.

Symmetric imputation. Following [Smith et al. \(2022\)](#): when an observed forward edge points to a removed node, a reverse edge is assumed to exist. For directed networks, this assumes all ties are reciprocated. For undirected networks, this is the natural reconstruction.

Probabilistic imputation (directed networks only). Following [Smith et al. \(2022\)](#): reverse edges to removed nodes are added with probability equal to the global reciprocity rate computed from respondent-responder dyads in G_{obs} . Repeated 100 times per missingness realization, with point estimates summarized as means across the 100 imputations.

Random augmentation bootstrap (Chapter 4 baseline). Following [Bellutta \(2026\)](#)’s Chapter 4: edge weights are added uniformly at random to G_{obs} until the total weight reaches an estimated true level. Repeated $B = 1,000$ times with credible intervals computed from the resulting distribution. This is the simplest bootstrap baseline and represents the prior bootstrap approach the framework is designed to improve upon.

Model-based imputation methods (simple and complex ERGM-based) are deliberately excluded from the comparison. [Smith et al. \(2022\)](#) established their performance characteristics on the same general class of networks; their computational cost scales poorly to networks the size of Balikpapan; and the framework’s positioning is as a scalable alternative to model-based imputation rather than as a competitor to it.

9 Measures and Evaluation Criteria

The validation evaluates each method on the same network measures used in [Smith et al. \(2022\)](#), allowing direct comparability with that prior work. Evaluation is conducted on two tracks: point-estimate accuracy

(comparable to prior imputation studies) and credible-interval calibration (the framework’s distinctive contribution).

9.1 Measures

The Phase 1 measure set is locked to the battery below. Each measure is computed in a weight-aware manner: for weighted networks the degree family uses strength (weight-summed degree) and the path-based measures (closeness, betweenness, mean inverse distance) use tie-strength edge semantics (Dijkstra on $1/w$, so stronger ties are shorter); for unweighted networks the degree family is the binary degree and the path-based measures use the binary (:ignore) semantics that reproduce [Smith et al. \(2022\)](#)’s convention. Bonacich power is already weight-aware and is computed identically in both cases. Global transitivity is computed by a direct $O(\sum_v d_v^2)$ triangle-counting algorithm in Phase 1 rather than by the triad census used for the Phase 0 characterization tables; the two agree on small networks but the direct algorithm scales to Marvel.

Node-level measures (centrality).

- In-degree / in-strength centrality (directed networks only)
- Out-degree / out-strength centrality (directed networks only)
- Symmetric degree / total strength centrality
- Closeness centrality (tie-strength weighted; binary for unweighted)
- Betweenness centrality (tie-strength weighted; binary for unweighted)
- Bonacich power centrality

Network-level measures (centralization).

- In-degree / in-strength centralization (directed networks only)
- Out-degree / out-strength centralization (directed networks only)
- Symmetric degree / total strength centralization
- Closeness centralization
- Betweenness centralization
- Bonacich power centralization

Topology measures.

- Largest component size (connected / weakly connected component)
- Largest bicomponent size (biconnected component)
- Mean inverse geodesic distance (tie-strength weighted; binary for unweighted)
- Global transitivity (direct triangle-counting algorithm)

Measures excluded from Phase 1. Two measures used in the Phase 0 characterization are deliberately *not* carried into the Phase 1 validation grid. The ranked-cluster tau statistic (Tau_{RC}) is a Phase 0 descriptive only: it requires a Monte Carlo null model per evaluation, which is prohibitively expensive at the scale of the validation grid, and it is undefined for undirected networks. Structural-equivalence blockmodeling is likewise excluded: its per-replicate cost does not reduce to the scalar-per-network form the validation grid requires. Both remain available as corpus-characterization tools but are out of scope for the degraded-network evaluation.

9.2 Track A: Point-Estimate Accuracy

For direct comparability with [Smith et al. \(2022\)](#), point estimates are evaluated using their established metrics:

Node-level measures. For each method and each missingness condition, compute the correlation between the true node-level centrality scores (computed on T) and the observed/imputed scores (computed on G_{obs} after each method’s reconstruction). High correlation indicates the method preserves the relative ranking of nodes. For the framework, the point estimate is taken as the posterior median across the B replicates.

Network-level measures. For each method and each condition, compute the standardized absolute bias:

$$\text{Bias} = \left| \frac{m(G_{\text{obs}}^{\text{reconstructed}}) - m(T)}{m(T)} \right|.$$

For the framework, the point estimate is again the posterior median.

9.3 Track B: Credible-Interval Calibration

The framework’s distinctive output is a posterior sample, summarized as a credible interval. Two evaluation metrics apply only to the framework (and to random augmentation, which also produces intervals):

Coverage. For each condition, the proportion of iterations in which the 95% credible interval contains the true measure value computed on T . Nominal coverage is 0.95. Coverage substantially below 0.95 indicates the framework’s intervals are too narrow (overconfident); coverage near 1.00 indicates the intervals are too wide to be informative.

Width. The mean width of the 95% credible interval across iterations, normalized by an appropriate scale (the standard deviation of the true measure across nodes for node-level measures; the absolute value of the true measure for network-level measures). Narrower intervals are more useful provided coverage is adequate.

9.4 Track C: Calibration Sensitivity

The framework’s calibration is also evaluated as the user-supplied ρ deviates from the induced ρ . Specifically, coverage and width are computed for each of the three prior conditions (correct, MCAR-assuming, opposite-sign) and compared. Graceful degradation — where coverage drops modestly and width increases moderately as the prior departs from truth — is a desired property. Catastrophic failure — where coverage drops to near zero under the wrong prior — would be a significant problem.

10 Success Conditions

The framework will be considered validated and ready for publication if the following conditions are met across the validation corpus.

10.1 Primary Conditions (must be met)

C1: Coverage under correct prior. Across all networks, induced ρ values, and missingness rates up to 30%, the framework’s 95% credible intervals achieve at least 90% empirical coverage when the user supplies the correct ρ . (The 90% threshold rather than 95% acknowledges that the framework is an approximate Bayesian procedure, not a strict one.)

C2: Point estimates competitive with simple imputation. For directed networks at moderate missingness (10–25%), the framework’s posterior median achieves bias and node-centrality correlation comparable to (within 10% of) the best-performing simple imputation method (probabilistic or symmetric, as appropriate) on each measure.

C3: Improvement over listwise deletion. Across all conditions, the framework’s coverage and width are strictly better than listwise deletion combined with naive bootstrap (i.e., the random augmentation baseline) when the missingness mechanism is non-MCAR (induced $|\rho| \geq 0.25$).

C4: Scaling to Balikpapan. The framework completes a single validation iteration on Balikpapan ($N = 1,347$) in under 10 minutes of wall-clock time on a single modern CPU core, making the full validation grid computationally feasible.

C5: Unweighted network support. The framework operates correctly on all four unweighted classes (unweighted directed, unweighted undirected) across the corpus’s unweighted networks. “Operates correctly” means: (a) the framework runs to completion without errors on these networks; (b) Stage 2 is correctly skipped (no edge-weight reversal attempted); (c) credible interval coverage under correct prior meets the C1 threshold (90%) on these networks. This condition formalizes the framework’s distinctive ability to handle unweighted networks, which the Chapter 4 framework could not.

10.2 Secondary Conditions (desired but not strictly required)

C6: Coverage under MCAR-assuming prior. When the user supplies $\rho = 0$ but the true induced ρ is ± 0.25 , coverage drops by no more than 10 percentage points relative to the correct-prior case. This indicates the framework is robust to mild prior misspecification.

C7: Graceful degradation under opposite-sign prior. When the user supplies a prior with the wrong sign, coverage may drop substantially but does not fall below 50% for any network and missingness rate up to 30%. This indicates the framework fails gracefully rather than catastrophically.

C8: E/I binning improves performance on betweenness. For networks with non-trivial community structure (modularity > 0.30), the framework with broker detection enabled achieves better coverage on betweenness centrality measures than a hypothetical degree-only variant.

10.3 Failure Conditions (would prompt revision)

F1: Coverage failure under correct prior. If C1 is not met for more than two networks in the corpus, the framework’s coverage mechanism (logistic bin-assignment, Laplace smoothing, or sampling distributions) requires revision before publication.

F2: Catastrophic failure under wrong prior. If C7 is violated — coverage falls below 50% with an opposite-sign prior at moderate missingness — the framework’s design is too sensitive to the prior and may not be suitable for general use.

F3: Failure to scale. If C4 is not met, the framework requires algorithmic optimization before validation can proceed at scale.

11 Validation Pipeline Architecture

The validation pipeline will be implemented in Python alongside the framework itself, following the module structure proposed in the accompanying implementation discussions. Key architectural decisions:

Data format. All networks are stored in a unified Parquet representation: a nodes table with node IDs and attributes, and an edges table with source, target, weight, and (for multiplex sources) layer identifier. This format is efficient for I/O, supports both Python and R/Julia analysis, and accommodates the eventual extension to multidimensional networks.

Reproducibility. All Monte Carlo iterations use seeded random number generators, with the seed determined deterministically from the condition specification (network, induced ρ , missingness rate, iteration index). This ensures the validation can be reproduced exactly and that individual conditions can be re-run without re-running the full grid.

Output. Framework runs produce per-iteration outputs in Parquet format: posterior samples for each measure, point estimates, credible interval bounds, and diagnostic information (realized correlation, binning mode used, etc.). Comparison method outputs are stored in parallel structures. Analysis and figure generation are performed in R using the `arrow` package to read the Parquet output.

Parallelization. The validation grid is embarrassingly parallel across (network, induced ρ , missingness rate, replicate) combinations. The Julia orchestrator (`build_degeneration_corpus`) uses `Threads.@threads :static` with a sampler-record cache that ensures both materialization mechanisms on the same cell share the underlying sampler call. The degeneration grid comprises approximately 72,000 records (the single-mechanism $16 \times 5 \times 6 \times 100 = 48,000$ records plus the second mechanism on directed networks only). At the Marvel-calibrated mean per-replicate cost of 0.147 seconds, the full doubled-mechanism degeneration grid completes in approximately 4 hours on a single workstation with sufficient threading. The downstream framework-evaluation grid — where each degeneration cell is multiplied by user-supplied ρ variations and B posterior replicates per measure — is correspondingly larger and is budgeted separately at the Track-A/B/C planning stage.

12 Reporting Plan

The final validation report will be a separate document, structured around the success conditions:

1. Corpus characterization (Phase 0 results, descriptive statistics for all sixteen networks)
2. Performance under correct prior (C1, C3): coverage, width, bias, and node-correlation results across the full grid
3. Comparison with simple imputation (C2): head-to-head point-estimate comparison
4. Calibration sensitivity (C6, C7): performance under MCAR-assuming and opposite-sign priors
5. Special analyses (C5, C8): unweighted network support and the value of E/I binning; the value of Stage 0.5 imputation when nominated non-respondents are identified
6. Scaling and computational performance (C4)
7. Discussion and limitations

Figures will follow the visual conventions of [Smith et al. \(2022\)](#): small-multiples decision-tree style figures showing the best method by condition, supplemented by coverage and width plots across the missingness-rate axis.

13 Risks and Contingencies

Risk 1: Framework coverage is too low under correct prior. If C1 fails systematically, the most likely causes are (a) the bin-assignment distribution does not produce the intended Kendall τ_b in practice (the analytic formula has been verified, but its match with realized sampling may diverge in edge cases), (b) the rank-rank matrix is too smoothed (overly Laplace-regularized), or (c) the Poisson weight model is wrong for some networks. Diagnostic outputs help identify which is the issue: per-replicate nominal vs. realized ρ , β -solver status distribution (especially :ceiling_hit rates), feasibility check output (whether the user’s ρ was structurally achievable on the network at the rate), and the per-cell degeneracy profile. A high :ceiling_hit rate accompanied by low coverage would point to feasibility issues; clean convergence with low coverage would point to the binning or weighting model. Mitigations include refining the β -solving procedure, reducing smoothing, or adopting weight distributions tuned to the input network.

Risk 2: Computational cost exceeds the budget. If the full validation grid takes longer than anticipated, the first cut is to reduce iterations per cell from 100 to 50 (raises the standard error on coverage estimates from ≈ 0.022 to ≈ 0.031 but remains usable for the primary conditions). The second cut is to reduce missingness rates from six to four (drop 5% and 50%, keeping 10%, 15%, 25%, 40%). The third cut is to reduce posterior samples from $B = 1,000$ to $B = 500$. The per-cell degeneracy profile (Section 3.5 sampler design) provides a fourth, targeted lever: cells that degenerate at high frequency need few replicates, so replication can be trimmed cell-by-cell rather than uniformly.

Risk 3: Balikpapan’s structure is anomalous. If Phase 0 reveals that Balikpapan has structural properties that make it an outlier in the corpus (e.g., extremely heavy-tailed degree distribution, multiple disconnected components), the validation may need to treat Balikpapan as a case study rather than including it in the main factorial. The reporting structure accommodates this contingency.

Risk 4: Empirical centralization values cluster too tightly. If Phase 0 reveals that all five empirical networks have similar centralization, the synthetic networks become the primary source of centralization variation. The synthetic design has flexibility built in to address this if needed.

14 Milestones

The validation is organized into seven milestones. Each milestone is expressed as one or more *necessary conditions for completion* — empirical or technical facts that must hold before subsequent work can responsibly proceed. The conditions are deliberately falsifiable: if the supporting evidence does not establish the condition, the milestone is not met, and the work either iterates or halts pending revision.

Milestone 1: Data Infrastructure

Necessary conditions.

- All five empirical networks load into the unified Parquet format without loss of information.
- Loaded networks reproduce published descriptive statistics (node count, edge count, density, direction) within rounding tolerance.
- The unified format accommodates all sources (five empirical, plus synthetic networks generated in Milestone 3) without source-specific schema variations.

Artifact. `nodes.parquet` and `edges.parquet` files for each network, plus a loader-validation report comparing loaded statistics against source documentation.

Gates. No subsequent milestone can proceed until the data is verified loaded; corruption or schema mismatch discovered later contaminates everything downstream.

Milestone 2: Empirical Corpus Characterization (Phase 0)

Status: complete. The eight Phase 0 tables (Tables 2–9) report the full characterization, and the conditions below are met.

Necessary conditions.

- Descriptive statistics from Section 3.5 (centralization, transitivity, mean degree, weight distribution, triad census, etc.) are computed for all five empirical networks and reported in the Phase 0 tables.
- The empirical corpus collectively spans enough structural variation to make the validation meaningful — in particular, at least two distinct centralization regimes must be represented across the empirical networks. (Met: degree centralization ranges from the near-uniform small networks to the strongly hub-skewed Balikan and Marvel cases.)
- Any empirical network that turns out to be structurally anomalous (e.g., disconnected, trivially small communities, degenerate degree distribution) is identified and a remediation decision is made (include with caveats, treat as case study, or exclude). (Met: Marvel’s largest component spans 70% of nodes; it is included with this fragmentation noted.)

Artifact. Phase 0 report: the six descriptive tables in Section 3.5, an empirical analog of Smith et al. (2022)’s Table 1 extended with weighted and triad-census characterizations, plus the narrative interpretation of structural coverage in that section.

Gates. The Phase 0 report determined that the synthetic network designs did not require redesign; the empirical corpus, with Marvel added at the large-clustered extreme, spans the intended structural range and the synthetic generators anchor the controlled centralization endpoints as designed.

Milestone 3: Synthetic Network Generation and Unweighted Variants

Status: complete. All synthetic networks and unweighted threshold variants are generated and present in the corpus (Table 1).

Necessary conditions.

- Synthetic 1 and Synthetic 2 are generated in both weighted and unweighted variants, each in directed and undirected forms (eight synthetic networks total).
- Thresholded unweighted variants of Moreno, Toledo, and Marvel are produced from the loaded weighted versions.
- Each synthetic network’s actual structural properties (computed post-generation) fall within the target ranges specified in Section 3.2 or are explicitly noted as deviations with justification.
- Together, the empirical and synthetic networks span low, moderate, and high centralization, both small ($N \leq 250$) and larger ($N \geq 600$) sizes, and all four cells of the weighted $\times$ directed design with at least three networks per cell.

Artifact. Synthetic and unweighted-variant network files in the unified Parquet format, plus a generation report comparing target and actual structural properties.

Gates. The full corpus must be finalized before the framework implementation enters validation testing, so that testing covers the network suite that will be used in production validation.

Milestone 4: Framework Implementation

Necessary conditions.

- All algorithmic components specified in the framework specification are implemented: bin assignment, P and R matrix computation, Louvain community detection with E/I scoring, β -solving for ρ -conditional bin distributions (against Kendall τ_b via the closed-form analytic formula), feasibility check (`feasible_rho_range`), automatic bin-count selection (`find_optimal_K` with `K = :auto`), Stage 0.5 imputation, Stage 1 node augmentation, Stage 2 edge-weight reversal, posterior aggregation.
- Unit tests cover each component with known-output cases, including edge cases (unweighted networks, $\pi_{\text{node}} = 0$, $\pi_{\text{edge}} = 0$, single-community fallback, sparse cell handling, structurally infeasible ρ , K -dependent ceiling-hit behavior).
- End-to-end smoke test on a small network (Moreno) produces output that is internally consistent: realized Kendall τ_b correlations between bin assignment and missingness across replicates match the requested ρ within tolerance (or report `:ceiling_hit` when structurally infeasible); bin counts match N_{add} in expectation.
- Computational performance permits at least one full framework run on Balikpapan ($N = 1,347$) in under 10 minutes wall-clock on a single modern CPU core (condition C4 from Section 6.1).

Artifact. Implemented framework as a Python package, unit test suite, and a smoke-test report demonstrating end-to-end operation and the Balikpapan timing benchmark.

Gates. Validation cannot run on an unimplemented or unverified framework. The Balikpapan timing constraint must be met before committing to the full validation grid; if not met, optimization work intervenes before Milestone 5.

Milestone 5: Comparison Baseline Implementation

Necessary conditions.

- Listwise deletion, asymmetric, symmetric, and probabilistic imputation methods are implemented and produce results matching [Smith et al. \(2022\)](#)’s reported behavior on Moreno when supplied with equivalent missingness conditions.
- Random augmentation bootstrap (from [Bellutta \(2026\)](#), Chapter 4) is implemented and produces results matching the Chapter 4 description.
- All baseline methods accept the same network and missingness inputs as the framework, producing comparable outputs (point estimates for all; credible intervals for random augmentation and the framework).

Artifact. Implemented baseline methods as part of the validation pipeline, plus a baseline-verification report showing that each baseline reproduces published behavior on at least one network.

Gates. The validation cannot make valid comparisons against baselines that have not been verified to behave correctly. Verification against published results is the necessary check.

Milestone 6: Pilot Validation

Necessary conditions.

- The full validation pipeline (network loading, missingness induction, framework execution, baseline execution, metric computation, output aggregation) runs end-to-end on Moreno across all five induced ρ values, all six missingness rates, and all three user-supplied ρ conditions, with reduced iterations ($n = 50$) for tractability.
- Pilot results are sane: coverage is in a plausible range (not 0 or 1 everywhere), interval widths scale reasonably with missingness, and method comparisons produce expected ordering on at least one condition (e.g., listwise deletion bias increases with missingness rate).
- No catastrophic failures are observed: no crashes, no NaN outputs, no negative interval widths, no coverage outside $[0, 1]$.

Artifact. Pilot validation report on Moreno: coverage and width tables, point-estimate comparisons, and an explicit assessment of whether the pipeline is producing sensible results.

Gates. The pilot exists specifically to catch problems before committing computational resources to the full grid. If pilot results are not sane, the full validation does not proceed; the issue is diagnosed and the pipeline revised.

Milestone 7: Full Validation and Reporting

Necessary conditions.

- The full factorial validation grid completes across all sixteen networks, five induced ρ values, six missingness rates, three user-supplied ρ conditions, and up to 100 iterations per cell (fewer where the degeneracy profile or Marvel’s per-replicate cost warrants).
- All measures from Section 5.1 are computed for the framework, all applicable baselines, and the true network values.
- Output is stored in the agreed Parquet schema with sufficient metadata to reconstruct any individual condition’s run.
- The primary success conditions (C1–C5 from Section 6.1) are evaluated against the empirical results, with each condition’s status (met / partially met / not met) documented.
- The secondary conditions (C6–C8) and failure conditions (F1–F3) are evaluated and documented.
- A validation report is drafted following the reporting plan in Section 8, with figures generated from the Parquet output.

Artifact. Complete validation dataset in Parquet format, plus a validation report addressing every success condition and providing the comparison narrative against the baselines.

Gates. The framework is considered validated and ready for broader use only if the primary conditions (C1–C5) are met. If any primary condition fails, the failure analysis in Section 9 determines next steps (revision of framework, revision of validation, or limitation acknowledgment in publication).

Summary of Milestones and Dependencies

#	Milestone	Primary artifact	Gates the start of
1	Data infrastructure	Loaded Parquet files	M2, M3, M4
2	Empirical characterization (Phase 0)	Phase 0 report	M3
3	Synthetic network generation	Synthetic Parquet files	M6
4	Framework implementation	Python package + smoke test	M5, M6
5	Comparison baseline implementation	Verified baseline code	M6
6	Pilot validation	Pilot report on Moreno	M7
7	Full validation and reporting	Validation report	Publication / release

Table 10: Milestone dependency structure. Each milestone’s completion gates the start of one or more later milestones.

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